

Closing the Dispersion Gap in Neural Limit-Order-Book Simulators: Findings from the August 2026 Fano Investigation

Honglin Fu

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Abstract

We investigate why neural marked-temporal-point-process (MTPP) simulators of limit-order-book (LOB) order flow under-disperse — their simulated event counts exhibit Fano factors far below real markets — and what repairs the gap. Across six experimental arms on three Coinbase assets we find: (i) the bounded-rate SS2P2 head, as trained, is not reliably self-exciting (measured impulse responses are sign-inconsistent and saturated); (ii) one-step maximum likelihood systematically misallocates excitation kernel spectra away from dispersion-relevant timescales (three independent demonstrations); (iii) covering the measured scales with the training window (seq 1024 $\rightarrow$ 4096) converts the Fano profile from a seed lottery into a reproducible, near-real rise — the largest single improvement found; (iv) at equal training budget a dispersion/accuracy frontier exists *along the training trajectory of a single model*: dispersion peaks early, accuracy peaks late, and no likelihood-selected checkpoint attains both; and (v) the residual level offset is dominated by rate-ceiling saturation (33% of events at the cap), not by amplitude learning or closed-loop drift. We situate these results against Jain’s Compound-Hawkes programme, whose moments-based calibration targets cross-scale count covariances by construction, and derive a certified route (linear Hawkes ground with pinned rate and settable branching) for which dispersion is a parameter rather than an emergent hope.

1 Framing: two identities

For bucketed counts N_Δ over windows of width Δ , the Fano factor is $F(\Delta) = \text{Var}(N_\Delta)/\mathbb{E}[N_\Delta]$. Two identities organize the investigation.

Cox decomposition. Conditioning on the intensity path, with $\Lambda_\Delta = \int_\Delta \lambda dt$,

$$F(\Delta) = 1 + \frac{\text{Var}(\Lambda_\Delta)}{\mathbb{E}[\Lambda_\Delta]}. \quad (1)$$

All excess dispersion is variance of the integrated intensity. A post-hoc rescaling $\lambda \rightarrow \kappa\lambda$ moves $F - 1$ only linearly with the rate: once the mean is calibrated, dispersion is fully determined by the model’s relative intensity-fluctuation structure.

Branching form. For a subcritical linear Hawkes process with branching ratio n , cluster sizes S satisfy $\mathbb{E}[S] = (1 - n)^{-1}$ and $\mathbb{E}[S^2] = (1 - n)^{-3}$, whence

$$F(\infty) = \frac{\mathbb{E}[S^2]}{\mathbb{E}[S]} = \frac{1}{(1 - n)^2}. \quad (2)$$

The *rising* profile $F(\Delta)$ across Δ is the fingerprint of long-memory (power-law-like) intensity autocovariance; single-exponential kernels plateau.

Empirical targets. Real Coinbase order flow (7 days, Jan 2026): $F(1s) \approx 40\text{--}70$ rising to $F(50s) \approx 394$ (SOL), 1247 (ETH), 1463 (BTC), still rising at the window edge. Implied branching (lower bounds): $n \approx 0.948$ (SOL), 0.971 (ETH), 0.973 (BTC). By Eq. (2) the seemingly small difference $0.948 \rightarrow 0.973$ quadruples asymptotic dispersion.

2 Asset heterogeneity: a tick-regime contrast

	trades (7d)	book rows	rows/trade	rate (ev/s)	1024-ev window
BTC	41,675	52.9M	1268	40.6	~ 25 s
ETH	94,048	55.5M	590	42.8	~ 24 s
SOL	40,048	20.1M	501	22.3	~ 46 s

Table 1: Raw data volumes and stream speeds (Coinbase spot, 2026-01-01..07).

Event-type composition differs qualitatively: inside-spread (IS) limit orders are 52% of BTC events, 34% of ETH, but only 7% of SOL, whose flow is instead cancel-dominated (43% CO). BTC/ETH are *small-tick* assets in Bouchaud’s sense (price-priority undercutting wars inside a multi-tick spread—the most self-exciting flow); SOL is effectively *large-tick* (1-tick spread, queue-priority churn). This microstructural contrast, not only the rate difference, explains the criticality ordering and why every simulator finds SOL easiest; it matches Jain’s thesis (Ch. 3) where stylized facts scale systematically with relative tick size.

3 Mechanism: the bounded head is not reliably self-exciting

ss2P2 factorizes intensity into a softmin-bounded scalar rate $\lambda(t) = s \text{softplus}(c - \text{softplus}(c - w^\top h(u))) \leq s \text{softplus}(c)$ and rate-neutral softmax marks. Stability is a bound on the rate *level*; nothing constrains the sign or magnitude of the event $\rightarrow$ intensity feedback.

Measuring the literal expected-offspring count (inject one extra event into a real history; integrate the additional intensity mass over 20 s) across six trained checkpoints: the median response is ≈ 0 to *negative* (-8.7 to $+0.2$ across seeds), positive in only 27–56% of states, with sign flips across seeds. Post-event intensity is pinned at the ceiling in every window. The model’s residual dispersion arises from a stereotyped impulse–decay envelope, not compounding cascades; cascades cannot compound because the second event in a burst adds nothing the first did not.

4 Experimental arms

All arms use the `ma_cbse` protocol: 62-channel event streams, TBPTT, categorical marks, streaming next-event evaluation on every test event, validation-calibrated 600 s rollouts, 3 training seeds $\times$ 3 rollout seeds per asset.

4.1 Negative and diagnostic arms

- **Book-state conditioning** (`ss2p2-1ob`; six top-10-ladder summary features): prediction unchanged (BTC accuracy 0.4484 vs 0.4478); closed-loop simulation *worse* on volatility-structure facts. Conditioning on self-replayed book state becomes a smoothing feedback

path. Consistent with the world-agent literature (Coletta et al.), which likewise conditions only on low-dimensional summaries.

- **lgm** (linear Hawkes ground $\times$ ss2p2 marks on the shared backbone): $\lambda_k = (\mu_0 + \sum_m a_m S^m(t)) \text{softmax}(z_k(u))$ with exact branching $n = \sum_m a_m / \beta_m$ (hard-projected) and the *rate pin* $\mu_0 = R(1 - n)$, which makes the free-rollout mean rate exact by construction (calibration finds $\kappa = 1.01$, vs $\kappa \approx 0.52$ for ss2p2). Two methodological findings: (a) ground accumulators must cold-start at their stationary mean $\mathbb{E}[S^m] = R/\beta_m$ (zero cold-starts bias windowed validation against slow-memory solutions; best-model selection then freezes runs at epoch 1 — observed on all nine initial runs); (b) with the fix, marks reach near-parity (0.186 vs 0.192 on SOL) while the 8-parameter clock costs 0.68 nats of timing likelihood. Fano: correct rising shape, seed spread ± 2 at 50 s (vs ± 217 for ss2p2) — dispersion as a property of certified parameters — but low level under $n = 0.95$.
- **Typed kicks, wider kernel bank** (lgm-k; per-channel kicks w_k , $M=6$ log-spaced kernels, $n = 0.97$): prediction improved to best-in-class on SOL (0.1967), but MLE moved kernel mass to $\beta \approx 0.008$ (~ 2 -minute memory), pushing variance beyond the observation window; *F regressed*. Third independent demonstration that one-step likelihood misallocates the spectrum.
- **Dispersion-matching loss** (ss2p2-disp; match $1 + \text{Var}(\Lambda_\Delta)/\mathbb{E}[\Lambda_\Delta]$ to the batch-empirical Fano at $\{0.5, 2, 8\}$ s): pins the level at trained scales reproducibly, *flat beyond the grid*; teacher-forced variance matching is not closed-loop compounding. A raised cap ($c=12$) produced the documented near-critical calibration fragility on one seed.

4.2 The window-coverage result (headline)

At seq 1024, TBPTT gradients truncate at ~ 24 – 25 s on BTC/ETH — before the 20–50 s scales where their simulated profiles flatten; SOL’s ~ 46 s windows cover the range, and SOL was the only asset where real-like rises appeared reliably. Setting seq/stride = 4096 (~ 100 s windows) with an otherwise identical recipe:

BTC, per seed	seq 1024 (12 ep)	seq 4096 (12 ep)	real
$F(50\text{s}), s1/s2/s3$	29 / 50 / 361	901 / 270 / 744	1463
rise $F(50)/F(1)$	$\times 2.5$ – 13.9	$\times 15$ – 32	$\times 20.9$

Table 2: Window coverage transforms the BTC Fano profile: all three seeds rise steeply (best seed within $1.6\times$ of real at 50 s). ETH: 2/3 seeds transformed. SOL: unchanged (negative control — its windows already covered the measured scales).

4.3 Equal-budget controls: a dispersion/accuracy frontier along the training path

seq 4096 with matched stride quarters the updates per epoch; 48 epochs restores the baseline’s total update count exactly.

Finding. Along the MLE trajectory of a single model, dispersion peaks early and accuracy peaks late; no likelihood-selected checkpoint attains both. The prediction/simulation tension long observed *across* architectures exists *within one training run*. The dispersion loss under a tight cap

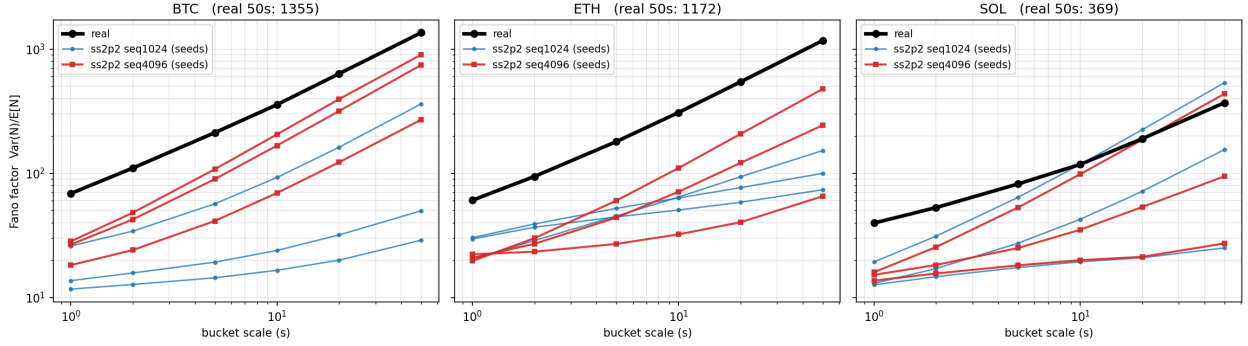


Figure 1: Per-seed Fano-vs-scale (log-log): real (black), seq-1024 seeds (blue), seq-4096 seeds (red).

BTC	accuracy	$F(1s)$	$F(50s)$ per seed
base (1024, 12 ep)	0.4478	17.1	29, 50, 361
w4k (4096, 12 ep)	0.4183	24.3	901, 270, 744
w4k48 (4096, 48 ep)	0.4514	16.6	232, 87, 230
w4kd (+disp. loss, 48 ep)	0.4519	11.1	52, 43, 43
real	—	70.0	1463

Table 3: Equal-updates controls on BTC (ETH/SOL analogous). Accuracy fully recovers — long windows are strictly better predictors at matched budget — but continued likelihood training walks the model away from the burst-faithful solution it passed through at 12 epochs.

converges all seeds to the smooth solution (best accuracy, worst dispersion), as predicted by the saturation diagnostic below.

4.4 Locating the residual offset

On the best w4k checkpoint (BTC): the dominating rate is 359 ev/s and the event-time intensity distribution has $p_{50} = 224$, $p_{90} = 357$, $p_{99} = 358$ — 32.8% of events occur above $0.9\times$ the ceiling. Teacher-forced on real histories, the model-implied F is only $1.2\text{--}1.4\times$ below real at every scale (e.g. 240 vs 290 at 50 s). The remaining level gap is therefore primarily *ceiling saturation* (which also kills the gradients of any dispersion objective), secondarily closed-loop drift; it is not an amplitude-learning failure. A moderate cap raise ($c=9$) on the equal-budget recipe is the decisive pending control.

5 The Jain reference point

Jain’s Compound-Hawkes programme (FRL 2024; thesis Ch. 4) never optimizes or reports the Fano factor; dispersion realism is scored by volatility signature plots and $|r|$ -autocorrelation, and his own signature plots undershoot the empirical level by $2\text{--}3\times$ — shape, not level, is the published state of the art. His dispersion derives from the estimator, not from likelihood: (i) kernels are fit by Kirchner-style binned-count least squares on a linear-log lag grid — an autoregression of current bin counts on lagged counts of every type, i.e. a moments fit whose target *is* the cross-scale count covariance; (ii) power-law kernels win AIC selection 100% of the time (straight lines over

six decades of lag); (iii) the kernel-norm matrix eigenvalue is capped at 1 by rescaling (a matrix `project_subcritical`); (iv) baselines are derived, not fitted: $\boldsymbol{\mu} = (\mathbb{I} - \mathbf{M})\boldsymbol{\Lambda}$ — the matrix form of the LGM rate pin; (v) compound order sizes (Dirac spikes at round numbers + geometric) matter enormously for price-domain dispersion (constant-size ablation wrong by $\sim 100\times$); (vi) a U-shaped time-of-day multiplier.

His RL chapter converts stylized facts into functional requirements for a market-making simulator, ranked by ablation evidence: intensity features (the clustering state) are the agent’s most valuable observable (removing them: Sharpe $+31.5 \rightarrow -20/-51$); relative queue position is next (removal yields manipulative or unprofitable policies); and impact-memory mis-specification *manufactures* strategies — exponential kernels admit dynamic arbitrage that the optimizer finds (“pump and dump”), rarely so under power-law kernels. Absence of dynamic arbitrage is effectively a required stylized fact for RL environments.

6 Conclusions

1. Prediction and simulation are different objectives; one-step MLE is blind to, and repeatedly misallocates, the kernel spectrum controlling $F(\Delta)$ — and (§4.3) trades dispersion away *within* a single training run.
2. The training window must cover the measured scales (§4.2): the cheapest large improvement found, converting the BTC profile from seed lottery to reproducible near-real rise.
3. The rate ceiling is the binding level constraint on the best current models (33% saturation); dispersion objectives are inert without headroom.
4. Certified dispersion requires linear-in-history structure: only the LGM ground offers a settable n , a pinned mean rate, and closed-form $F(\Delta)$ at any horizon. The Kirchner binned regression is the proven recipe for fitting it (typed, per asset) to cross-scale count covariances, post hoc, with the trained mark head untouched.
5. Targets must be per-asset: tick regime (IS share) drives criticality; n differences of 0.025 are $4\times$ Fano differences.
6. For the eventual market-making agent, the simulator acceptance checklist (after Jain Ch. 6) is: genuine clustering exposed as intensity state; honest queue mechanics; arbitrage-consistent impact with power-law memory; reactive spread/liquidity dynamics; realistic frictions.

Provenance. Experiments: `ma_cbse/{btc,eth,sol}/tags/ss2p2-full,ss2p2-lob,lgm,lgm-k,ss2p2-disp,ss2p2-w4k,ss2p2-w4k48,ss2p2-w4kd,ss2p2-w4kc9` (pending). Code: `honglinfu98/simulation` commits `7b97b21..88d22cc`; mirrors in `dmm_sim/`. Full lab log: `docs/fano_investigation_2026-08.md`.